

Exploring how Emotional Challenge and Affective Design in Games Relates to Player Reflection

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ABSTRACT

Emotional challenge contrasts with the common perspective of challenge in games. The latter focuses on performance, while the former focuses on presenting players with heavy themes that induce complex emotions. To explore how emotional challenge impacts the player experience, we examined the relationship between player reflection and affective game design patterns—conducting a survey with 53 respondents that prompted recalling an emotionally challenging situation in a video game. We then scored perceived emotional challenge ratings and coded the responses according to referenced affective game design patterns to categorize levels of reflection achieved. We found a significant difference ($p < 0.011$) in emotional challenge ratings and mean ranks increased across the five reflection groups, from lowest reflective level (non-reflective description) to highest (critical reflection). This demonstrates that the more emotionally challenged participants felt, the greater the depth of their reflection was. We also discovered the following design patterns to be prominently referenced in respective reflection levels: emotional decision-making, identification, and negative/uncomfortable feelings such as guilt.

CCS CONCEPTS

• Human-centered computing → User studies; Empirical studies in interaction design.

KEYWORDS

Challenge, Game Design, Reflection, Emotional Challenge, Affective Design, Player Experience

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1 INTRODUCTION

Player enjoyment and fun can no longer be simply gauged by the frequency of smiles [52] while playing a game. Technology has progressed to improve the levels of immersion in which games

tell complex and responsive stories that can emotionally impact players deeply [28, 56]. Isbister [28] argued that "games can actually play a powerful role in creating empathy and other strong, positive emotional experiences." With digital games in particular, Karpouzis & Yannakakis [56] said that we now have the "ability to sense player emotion, through cameras, microphones, physiological sensors and player behaviour within the game world, and utilise that information to adapt gameplay accordingly or generate content predicted to improve the player experience and make the game more engaging or interesting." Clearly, combining games with rising technology can make their emotional power effect even stronger when designed for thoughtfully.

The type of challenge presented to players in these emotionally-driven games is a bit different from what we are used to. In addition to what we typically know as hard fun (or performative challenge [19]), there is a type of fun closer to what has been coined as "serious fun" that depends on emotions and reflecting upon them during play [35]. Serious fun is when players "aim to change how they think, feel, behave, or to accomplish real work" [35].

Reflective player experiences have been proven to hold value for the gaming community, in that reflection is a process that players generally enjoy [33, 41]. This is particularly important in the context of games that address serious and/or complex themes in their narrative, seeking to induce emotional reactions from their players. "Emotional challenge" [17] is a term that describes and measures this phenomenon—the "hard fun" of emotional play—and researchers are increasingly curious about how it relates to player experience and reflection [17, 19, 41].

One motivation to study how to design for reflective player experience is because games are increasingly used in the area of education and games for learning, especially towards social-emotional learning [27, 29, 30, 48]. In learning processes, reflection about the content increases the depth of knowledge, contextualizes knowledge, and helps build structural connections in knowledge and social connections among learners [15]. Therefore, understanding how emotional experiences in games relate to reflection is increasingly important, especially for games that aim to impact players in deeper ways.

Even games that are aimed primarily for entertainment have unintended learning experiences that are influential towards players' real lives [49]. When players were asked to recount emotionally moving gaming experiences, they were able to describe moments that were experienced an entire decade ago but have still left a lasting emotional impact [54]. Games researchers and designers alike could benefit from research in this field to continue to create monumental video games.



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In this paper, we sought out to better understand emotional challenge in games and how to best design for it, focusing on how emotional challenge relates to meaningful reflective player experiences. The main research questions that our work seeks to answer are:

- How does emotional challenge in games relate to reflection?
- What affective game design patterns are important for reflective player experiences?

To answer our research questions, we surveyed 53 individuals that recalled an emotionally challenging situation in a video game—utilizing a qualitative coding process and nonparametric Kruskal Wallis H test for analysis. Our findings show that players do experience various levels of reflection dependent on the level of emotional challenge they experienced in a game. Specifically, there is a significant difference in emotional challenge ratings across the five reflection groups and mean ranks increased across the five reflection groups, from lowest reflective level (non-reflective description) to highest (critical reflection). We also uncovered the types of affective game design patterns most prominent in higher-quality reflections, i.e., autonomy-based and negatively-valenced patterns. We then conclude by discussing what our findings mean for game design with respect to reflective and emotional challenge.

2 RELATED WORK

2.1 Evolution of Stories in Games and Emotional Challenge

While there were narrative-based games such as *Oregon Trail* (1971) and *Colossal Cave Adventure* (1975), story was not at the forefront of the most popular video games such as *Pong* (1972) and *Space Invaders* (1978). Narrative in games has evolved from "player assumption, to written backstory, to written narrative context, and eventually to text, graphics, and movies integrated directly into the game experience" [26]. The popular perception of challenge in games was limited at the time to what was called functional [17] or performative challenge [19], which addressed players' physical limitations to interact with the game (i.e. the speed and accuracy with which actions can be performed) [19] or skill with strategy [17]. Game design therefore focused on flow [18] and perfecting just the right amount of performative challenge to maintain a player's interest and focus [31, 32].

Cole argued that the other side of the spectrum were games that focused on the game story and feelings that it induces in players; this was appropriately named "emotional challenge" [17]. Instead of challenge via physical game mechanics, the foundation of emotional challenge is built upon emotional and intellectual engagement with an unfolding narrative [26]. The challenge eventually became the task of measuring such an experience, similarly to how performative challenges are measured by enjoyment and flow [46, 50]. Denisova et al. [19] developed such a survey via Likert Scale method, which had both components of performative challenges and, most importantly in our case, emotional (scale items detailed in "Methods" section). Other previous work similarly explored measuring emotional challenge in relation to immersion, enjoyment, and competence [9], in 'core gamer' and avant-garde titles [17], as well as on the virtual reality (VR) platform [44, 45].

2.2 Reflective Player Experience

Reflection is a commonly desired outcome in HCI studies for its bridging of technology and the surrounding culture [4]. However, research into reflective design is still underexplored [47]. Reflection is tricky to research as there are various interpretations, so Fleck & Fitzpatrick addressed this issue by creating a framework to discuss the five levels of reflection [24]:

- (1) The first level is *non-reflective* or simply describing events "without further elaboration or explanation".
- (2) The second level is *reflective description*, which includes justification for actions but in a reportive way.
- (3) The third level is *dialogical reflection* which starts to incorporate piecing relationships between ideas and considering other points of view.
- (4) The fourth level is *transformative reflection*, which goes deeper to challenge one's views and include actual change in understanding or action.
- (5) The fifth and final level is *critical reflection*, which addresses wider ethical and societal considerations, and is the most rare type of reflection to obtain.

There is much to learn about reflection in games as it increases understanding of "engagement beyond the moment-to-moment experience of gameplay" and is considered a crucial part of learning [41]. Mekler et al. [41] found that while higher-level reflection in the above framework is difficult to achieve, players find the activity of reflection in itself valuable. Additionally, Whitby et al. [54] found that players also reflect during gameplay ("endo-transformative" behaviors) in addition to outside of it.

There appears to be a relationship between complex emotions and reflective player experiences [5, 42]. Bopp et al. [7] asked players to recall "emotionally moving" gaming experiences and found that players actually enjoyed experiencing negatively-valenced emotions such as sadness. Gowler & Iacovides [25] dove deeper into uncomfortable emotions in gameplay and found that it often served as a "catalyst for reflections on wider societal issues and concerns". Similarly, Whitby et al. [55] found a sequence of trigger, reflection, and transformation constituted perspective-challenging experiences. Clearly, there is value in learning how to design for these more complex gaming experiences, as they can lead to long-lasting, and even transformative, reflective impacts on players. The importance of reflection towards learning then suggests that games can be used to address complex themes and social-emotional learning [27].

2.3 Affective Patterns and Social-Emotional Learning

Generally speaking, the affective domain of learning consists of how we perceive, experience, and process the emotional aspects of social interaction [38]. While understanding and managing one's emotions are key affective elements, so is establishing relationships with others [40, 51]. One way to build and maintain relationships with others is through social-emotional learning, which consists of affective components such as self-awareness, social awareness, and relationship skills [43].

As stated earlier, with the rise of incorporating emotional challenges in games, a way to identify, evaluate, and create these challenges is through using affective patterns. Examples of affective patterns in games include, but are not limited to: 1) *avatar emotional expression*: "facial expressions alter in response to in-game events or player through player commands", 2) *avatar display of human frailty*: "inclusion of less than optimal physical traits in an avatar", and 3) *the traumatized avatar*: "an avatar with emotional trauma [that] allows players to gain a deeper understanding of what drives the avatar and thus empathize with it" [22].

Affective patterns in games can help promote social-emotional learning by prompting players to not only think about specific scenarios, but also to get a (virtual) first-hand *experience* with perspective-taking and decision-making [37, 53]. As a result, this could also induce various levels of reflection among players.

For example, consider this scenario where a player has to choose whether or not share their feelings to an in-game character. If the player does so, then their bond with the character will deepen, and they may be more likely to help the player in times of need amidst battles. However, if the player does not do so, then their bond may remain the same or weaken, as they will not be as inclined to interact and/or help the player when they are available. Regardless of the player's decision, this affective pattern provides opportunities that encourage player reflection.

3 METHODS

3.1 Study Goals

The goal of this study was to explore how emotional challenge in games relates to reflection and affective game design patterns. As previously discussed, the evolution of game narratives has enabled games to address complex emotional themes in innovative, personalized ways. Games now have the ability to induce various emotional challenges through both the game story and how it is presented to players via affective game design patterns. We believe that inducing emotional challenge through affective design patterns influences the depth of reflection one has about their gaming experience. Therefore, we expect that the more emotionally challenged players feel, the greater the depth of their reflection of the experience will be.

Additionally, our study also explored which affective design patterns were most prominent, as a means to determine how to best induce reflective emotional challenge through design. We also expect to see that responses, within each reflection level, will have similarly referenced affective game design patterns. Lastly, we anticipate that there will be a significant difference among the players' emotional challenge ratings for each reflection level group, signifying that each group has distinct range of emotionally challenging experiences.

3.2 Participant Recruitment

We recruited 59 participants through social media (e.g. Twitter, Facebook, Slack, etc.) using a combination of the authors' personal contacts and Amazon Mechanical Turk (MTurk). The only conditions for participation was that participants had played video games previously and responses needed to be in English.

3.3 Procedure

To obtain a wide diversity of responses, we created a Qualtrics survey that asked participants to recall an instance where they faced an emotionally challenging situation in a video game. As this method did not have as much flexibility as interviews, we provided a definition of "emotional challenge" as "challenge which confronts the player with emotionally salient material or the use of strong characters, and a captivating story. A player cannot overcome emotional challenge with skill or dexterity, but by resolving tension in the narrative, by assessing their identification with game characters, and by resolving ambiguities" [19]. Afterwards, we then structured the response section into these prompts:

- (1) "Recall a time you experienced an emotional challenge in a video game."
- (2) "What made that experience emotionally challenging?"

Participants were free to write, with as much or as little detail, about any game in their responses.

After describing their experience, participants scored it according to challenge ratings using the Challenge Originating from Recent Gameplay Interaction Scale (CORGIS) [19].

3.4 Measures

For our survey questions, we utilized the emotional challenge dimension of the CORGIS measurement tool—which has good internal consistency with Cronbach's α of 0.84 [19]. We only selected the emotional challenge dimension as it focused on our research questions. It asked for the degree of agreement with the following statements through a 7-point Likert scale ranging from strongly disagree (1) to strongly agree (7):

- This game is more than just a game to me
- The things that happened in the game made me sad
- I invested much thought into the game
- I felt a sense of responsibility for characters and events in the game
- The game made me think about real life issues
- Playing the game was stimulating
- I felt a sense of suspense when playing the game
- The game had moral dilemmas in it where the choice was not obvious
- The game involved making moral choices that I didn't agree with"

While CORGIS also measures other aspects of challenge such as performative challenge, those constructs were not relevant, as our study focused on emotional reflection, and therefore excluded from analysis.

3.5 Data Analysis

3.5.1 Qualitative Data Analysis. After collecting the data, we had two researchers take part in the analysis process. We utilized a deductive thematic analysis [12] approach, as we already had a set of affective patterns to look out for in the data. We merged the responses from our two prompts and coded them together as one unit. We independently deductively coded the written portion of the survey responses (53 total) for referenced affective game design

Table 1: Affective Patterns Codebook, from Dormann et al. [20–22], Bjork & Holopainen [6], & Lankoski & Bjork [34]

Affective Game Design Patterns	Definitions
Animal companion sims	A simulation of a real animal, intended to reproduce the effects of actual pet ownership.
Avatar display of human frailty	Describes the inclusion of less than optimal physical traits in an avatar. These traits more closely mirror the reality of human life.
Avatar emotional expression	Avatars’ facial expressions alter in response to in-game events or through player commands.
Call to action	Provides functions within the game framework that inspire players to step beyond the game and actively participate in social activism.
Competing for attention	The competition between several characters to get attention of one character.
Consequences of long ago actions	Reminds players that the current outcome is a consequence of a previously made decision.
Emotional decision making	Affecting player emotions through decision-making processes.
Emotional immersion	Being emotionally affected by the events that occur in a game.
Empowerment	Players feeling that they can affect the events and the final outcome of a game.
Gain allies	The goal to add new members to a social network defined as an alliance.
Healing/nurturing others	Refers to the use of game resources to heal damage sustained by teammates and/or NPC companions.
Identification	The characters or parts of the game with which players identify.
Internal conflict	Having a set of desirable goals where progress in one typically makes others more difficult.
Negative emotions (anger, guilt, shame)	Uncomfortable emotions that are triggered by gameplay or themes in a game.
NPCs with emotional masks	The NPC who uses an emotional mask is a character that displays one kind of emotion when they are secretly feeling another.
Sacrificial action	The action of prioritizing others’ well-being (NPCs and, in multiplayer games, other players) ahead of the player’s own character’s well-being and advancement.
Shared learning	The opportunity for players to share lessons from the game with other players. Through direct and indirect cooperation, players work together to master the game.
Social dilemma	Instantiated in a game through internal conflict.
Social maintenance	Perform actions to redefine and refine the relation to a group.
Sympathy for victims	Involves support for feeling concern and compassion for victims.
The traumatized avatar	An avatar with emotional trauma allows players to gain a deeper understanding of what drives the avatar and thus empathize with it.

patterns based from Dormann et al.’s previous works [20–22]. Examples of such patterns in addition to those previously mentioned include: healing/nurturing others, emotional decision-making, and sacrificial action [22] (see Table 1 below for the complete codebook).

To find the top affective game design patterns presented in Table 1, we calculated the percentage that a design pattern appeared out of all other patterns that surfaced in a reflection level.

Next, we coded individual responses according to their highest achieved level of reflection based on Fleck & Fitzpatrick’s [24] framework, which could be (from highest to lowest quality): critical reflection, transformative reflection, dialogical reflection, reflective description, and description. We selected this framework due to its usage in previous reflective player experience studies [41] for consistency. Two researchers conducted this analysis separately; for discrepancies in coding, a discussion afterwards was conducted

until consensus. A summary of coding themes and representative responses can be found in Table 2.

With two qualitative coders, the result of our interrater agreement analysis was substantial (mean Cohen's $\kappa = 0.794$; range $\kappa = 0.665$ to $\kappa = 0.91$). Since we used the coding reliability thematic analysis approach, interrater reliability is a key measure of coding quality [13, 14].

3.5.2 Quantitative Data Analysis. As we had a smaller sample size due to grouping participant responses by reflection level, we then used the nonparametric Kruskal Wallis H test to analyze our quantitative data, given one of our assumptions that *there is a significant difference among the aforementioned reflection groups regarding emotional challenge ratings—i.e., meaning that the more emotionally challenged players feel, the deeper their reflection about their gaming experience will be.* We then conducted post-hoc tests adjusted with Bonferroni correction to evaluate pairwise comparisons among the five reflection groups.

4 FINDINGS

4.1 Participants and Emotionally Challenging Games

Of the original 59 participants describing emotionally challenging gaming experiences, six had to be excluded due to responses not relating to a video game (i.e. real-life sports play). Within the remaining 53 responses, the age range from participants were from 18-66+ years old, with 28 who identified as male, 22 as female, and 3 as non-binary. Their years of experience playing video games ranged from 0 to 3 years to 44 years. The games referenced by more than one participant as an emotionally challenging experience were *The Last of Us* (n=3), *Red Dead Redemption 2* (n=2), and *Undertale* (n=2). Though we initially expected mostly singleplayer experiences due to the focus on story emotional challenge, competitive and cooperative multiplayer online games (n=14) such as, *League of Legends* and *World of Warcraft*, were a notable portion. The most popular game genres included Role-Playing Games (n=44), Action Adventure (n=41), and Shooters (n=33).

4.2 Emotional Challenge Ratings

Through the nonparametric Kruskal Wallis H test, we found there be a significant difference ($\chi^2(4, N = 53) = 13.108, p < 0.011$, effect size = .252) in emotional challenge ratings across the five reflection groups. Descriptive statistics can be found in Table 3 and a matrix of post-hoc pairwise comparisons in Table 4. Apart from the transformative reflection group (mean rank=37.08, N=6) which was slightly higher than critical reflection (mean rank=36.80, N=5), the emotional challenge ratings/mean ranks progressively increased from lowest to highest levels of reflection (also observable through the means). Observing the ranks, it is clear there is a large jump (nearly double) in emotional challenge ratings from reflective description (mean rank=22.78, N=18) to dialogical reflection (mean rank=35.45, N=10) and above levels.

Table 4 below displays the conducted post-hoc tests adjusted with Bonferroni correction to evaluate pairwise comparisons among the five reflection groups, in which the significant differences demonstrate that the reflection groups are distinct from one another.

4.3 Levels of Reflection

4.3.1 Non-reflective Description and Reflective Description. The first two levels of reflection consist mainly of statements that describe what happened during the participant's gaming experience, i.e. mostly dealing with the emotion unrelated to the game's narrative. What differentiated reflective from non-reflective is simply whether the response included reasons for actions or interpretation, but in a reportive way [24].

For instance, in responses #1 and #2 from Table 2, while the participants describe feeling frustrated, the reflection repeatedly focused on specific functional/mechanical aspects of gameplay, such as time, logistics, and strategy that impacted their experience. While there was *Emotional Immersion* in the response, it was largely due to the *Anger/Frustration* of losing.

4.3.2 Dialogical Reflection. As discussed earlier, dialogical reflection is a step deeper into reflection by examining relationships. In response #3 from Table 2, it is clearly a deeper emotional reflection than the last responses, as it actually involves identification with events happening in the game and a perspective taken towards a lesson learned about emotionally coping.

4.3.3 Transformative Reflection. Response #4 involves a participant who similarly goes through a process of perspective-taking, but with a focus on *Emotional Decision-Making*. They described struggling with the death of their main character, as choices they made led to this ending caused by having "high honor" in the game, seeing the impact of *Consequences of Long Ago Actions*. They end by saying that the experience has taught them that doing "the right thing doesn't always lead to a happy ending", so they need to practice being more present in enjoying precious moments in real life.

4.3.4 Critical Reflection. While there were not many responses that achieved critical reflection due to its required depth, these responses represented the highest quality reflections we obtained. Response #5 was such an example of critical reflection. The participant emphasized that their experience required a lot of *Emotional Decision-Making* that brought about moral/*Social Dilemmas* and *Sacrificial Action*, due to there being *Consequences of Long Ago Actions*. The way they deeply pondered over their decisions before making them meant they felt *Emotional Immersion* while they had *Empowerment*/agency over those decisions. Because the participant included a reflection on ethics and morals and expressed that they carried over this experience into real life outside of the game, the response was coded as critical.

4.4 Affective Game Design Patterns

As with the above examples, each response was coded according to referenced affective game design patterns. Figure 1 below displays the top game design patterns for each reflection level. The deeper/higher levels of reflection (e.g. critical reflection) also contain the patterns in subsequently lower levels.

Table 2: Example used for scoring qualitative answers and corresponding representative responses from study participants.

Representative Response	Reflection Level	Affective Patterns
1) "I recall when Animal Crossing came out in 2020, I was very much addicted to building my own island. However it takes a lot of hours just to travel around other islands to retrieve resources. I was constantly stuck not knowing the correct steps to take in order to speed things up."	Non-Reflective Description	N/A
2) "I know I learned to accept that I can't win and walk away until I'm ready to try again. I don't know if I would consider that lesson coping, though. As I play, I still get frustrated and catch myself being explicitly aggressive (many WTFs and are you serious go on). I think the coping may be that every time I try to fight a boss after I die, I come up with a different strategy that might work."	Reflective Description	Emotional immersion, anger
3) "I learned that there are times when you are going to have to do things that are tough. As someone in the military, I have to deal with that quite often. So to me I learn to separate my feelings from the things that I have to do that will be mentally tough on me."	Dialogical Reflection	Emotional immersion, identification, emotional decision-making, empowerment
4) "I think playing the ending of the game <i>Red Dead Redemption 2</i> was quite an emotional challenge. By the time you reached the end, you really felt heavily invested in the main character Arthur Morgan. There's an intense gunfight right before a pivotal moment in the game. We find out that Arthur dies because of injuries and disease. It's especially emotional because you can only get this ending if you have high honor, which means you made very noble choices throughout the game ... I felt emotionally attached to the character. My actions shaped the ending. I learned that doing the right thing doesn't always lead to a happy ending. Made me think that I have to live in the moment and enjoy life while I can."	Transformative Reflection	Emotional immersion, emotional decision-making, empowerment, consequences of long ago actions
5) "Pretty much every decision in Dragon Age Origins. ...All of the experiences were moral dilemmas. There was no clear "right" answer, but potentially a "best" answer depending on one's own view of ethics/morals. For some of the decisions involving party members, there's a sense of connection to them and the fear that making the "wrong" choice will harm them or disappoint them or cause them to leave the party or end the relationship with my character. Basically, moral dilemmas with difficult consequences. Thinking through each situation and the potential courses of action reinforces the way that I think through ethical dilemmas IRL. So I guess it was maybe good practice in managing emotions/stress associated with decisions in a no-risk environment."	Critical Reflection	Emotional immersion, identification, emotional decision-making, empowerment, consequences of long ago actions, social dilemmas

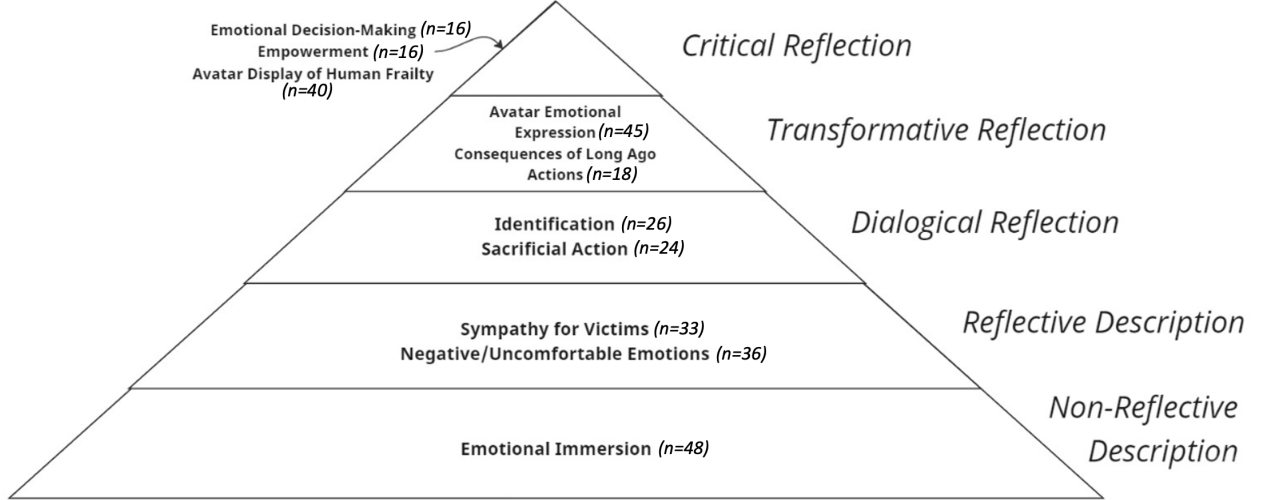
Table 3: Emotional Challenge Ratings among Reflection Groups

Reflection Level	N	Mean Rank	Mean	Standard Deviation
Non-reflective Description	14	18.57	4.8214	0.67957
Reflective Description	18	22.78	5.0389	0.84376
Dialogical Reflection	10	35.45	5.72	0.75836
Transformative Reflection	6	37.08	5.8	0.56214
Critical Reflection	5	36.80	5.84	1.04547

Table 4: Matrix of Post-hoc Tests with Bonferroni Correction for Pairwise Comparisons across Reflection Groups

Non-reflective Description (ND)	1.0				
Reflective Description (RD)	0.444	1.0			
Dialogical Reflection (DR)	0.008	0.037	1.0		
Transformative Reflection (TR)	0.014	0.049	0.838	1.0	
Critical Reflection (CR)	0.023	0.072	0.873	0.976	1.0
Reflection Level	ND	RD	DR	TR	CR

N=53

**Figure 1: Top Affective Game Design Patterns per Reflection Level**

5 DISCUSSION

5.1 Emotional Challenge and Levels of Reflection

We found that there is a high significance in differences for emotional challenge ratings among the five reflection groups, i.e., critical reflection, transformative reflection, dialogical reflection, reflective description, and description (from highest to lowest quality) [24]. Upon closer inspection, we noted that the lowest level of reflection, *Non-reflective Description*, has the lowest mean rank of 18.57. It is in the dialogical reflection level that the scores for emotional challenge ratings greatly increase. According to Fleck & Fitzpatrick, dialogical reflection involves "searching for relationships between ideas and experiences in order to generalize from them and reach a different level of understanding" [24]. Therefore, it is at this level that thought processes shift from general descriptives to deeper reflection. This supports our assumptions that there would be a significant difference among the above reflection groups with respect to emotional challenge ratings and that they would increase from lowest to highest levels of reflection. These findings demonstrate that it can be inferred that *the more emotionally challenged participants felt about their gaming experience, the greater the depth of their reflection about it was*. This is crucial in furthering our understanding of reflective player experiences and how to better design for them, as it has been demonstrated from recent studies that video

games do provoke complex emotions [9] and reflective experiences [41] that players enjoy.

It should be noted that we had somewhat different findings from Mekler et al. [41], in that they uncovered little to no instances of higher-level transformation and critical reflection, while we did uncover several instances. Upon further inspection, we believe that our different approaches to asking players to reflect had an influence on their varying responses. To reiterate, we gave a definition for "emotional challenge", whereas Mekler et al. [41] refrained from doing so and kept it broad. For our purposes, we chose to give a definition to ensure that participants had a baseline understanding of what we meant by emotional challenge since that is what we were measuring in our study. We do understand that this could have influenced the way players reflected; however, given that we still had a disproportionate amount of responses in the lower-level dialogical reflection and below groups, we do not believe that this skewed our results too much. Nevertheless, we do argue that our findings support Mekler et al.'s [41] in that deeper levels of reflection from a game seems to require willingness from the player to think deeper and that may also require a trigger from within the game's design.

5.2 Affective Learning Game Design Patterns

Design knowledge on stimulating or induce deeper reflection through game design is still largely unexplored [30]. While ultimately, the

levels of reflection do share affective game design patterns among each other, it is beneficial for game designers and researchers to know which patterns are useful for eliciting deeper thoughts and conversations among their players when designing meaningful play.

Figure 1 shows the top affective game design patterns for each reflection level. Our results demonstrate that *Emotional Immersion* is at the core of emotionally challenging player experiences, as it was present from Non-reflective Description all the way to Critical Reflection. This is because one must first be emotionally immersed and present in order to feel emotionally challenged at all [2]. However, to achieve higher levels of reflection, it requires layers of emotional immersion, as broken down in the next subsections.

5.2.1 Empathy and The Value of Negative/Uncomfortable Emotions. Our findings support previous work that negative emotions lead to positive, meaningful experiences which players do enjoy [7, 41]. They also support Marsh & Costello's [39] previous work that found unease or discomfort provoked interpretative reflection in players. They also argued that "lingering" emotions post-encounter for players are important to encourage reflection. When transitioning from non-reflective to *Reflective Description*, the most prominent affective game design patterns were *Sympathy for Victims* and other *Negative/uncomfortable Emotions*, such as guilt or sadness. Diving into deeper reflection levels clearly requires some type of trigger, such as inducing some mental or emotional discomfort in players.

Sympathy, guilt, and sadness are emotions that are related to empathy [23]. While a controversial topic, empathic processes can be deduced to three major subprocesses: emotional simulation, cognitive/perspective-taking, and emotion-regulation [23]. This first step into reflection with empathetic emotions complements that initial emotional simulation and immersion that happens in the beginning of empathic processes. Therefore, it makes sense that the pattern for *Negative/uncomfortable emotions* first start to appear at the lower *Reflective Description* level, as this pattern acts as that initial trigger into deeper thinking for players.

Recall that there is a large difference in our findings for emotional challenge ratings from *Reflective Description* (mean rank=22.78, N=18) to *Dialogical Reflection* (mean rank=35.45, N=10). At the *Dialogical Reflection* level, the game design patterns for *Identification* and *Sacrificial Action* become more eminent. Hence, we argue that these patterns complement the cognitive/perspective-taking subprocess in empathic processes. *Identification* is linking or relating oneself to someone else or something in a game – whether that is the player-controlled character (PC), a non-player controlled character (NPC), or events that have occurred. *Sacrificial Action* is "prioritizing others' well being... ahead of the player's own character's well-being" [22]. Both affective game design patterns require perspective-taking to relate to aspects of a game and thus is why they appeared more frequently at higher level of reflections (dialogical reflection and above).

Similarly, the patterns *Avatar Emotional Expression* and *Avatar Display of Human Frailty* appeared often in the highest levels of reflection. The former describes facial expressions being responsive to game events or player actions, while the latter describes including "less than optimal physical and mental traits in an avatar... [to] more

closely mirror the reality of human life" [22]. These patterns serve more as stimuli to which players may react emotionally.

We then argue and support others' findings [7] that inducing difficult emotions do not have to be shied away from. Psychological findings in fact have argued that "bad is stronger than good" [3], and that "negative" experiences often have greater impact than positive ones. This is not to say that positive game design should be avoided. Our findings simply highlight that to create emotionally challenging experiences in games, stimulating deeper reflection of negative/uncomfortable emotions through affective game design patterns should not be ignored.

5.2.2 Identification in Games. Recall that identification is a pattern that started to appear more at the *Dialogical Reflection* level. Additionally, 26 out of 53 of games referenced were actually Role-Playing Games (RPGs), in which identification is a huge part of gameplay. According to Cohen [16], identification can be defined as a "process that consists of increasing loss of self-awareness and its temporary replacement with heightened emotional and cognitive connections with a character". In addition to a loss of self-awareness, identification also consists of "textual features intended to provoke identification" and "internalizing a point of view rather than projecting one's own identity". Since we did not specifically ask participants whether they were self-aware throughout their gameplay experience, we cannot make confirmations to this point of the identification process. However, we can for the latter two parts, as "textual features" can be equated to design elements in games (i.e. character, quest, and narrative designs) and "internalizing a point of view" can be equated to the five levels of reflection (from descriptive to critical). Both of which can be obtained from the survey responses directly and contribute to experiencing emotional challenges in games. Therefore, it supports the fact that the *Identification* pattern becomes more prominent at the *Dialogical Reflection* level where players start forming connections between pieces of knowledge that they have gained. The connection between the in-game and out-of-game reality are forged when players begin to identify with characters and events happening in the game.

For example, consider this survey response when talking about a specific emotional challenge from *NieR: Automata*:

"...one of the bosses was a strong, talented, and passionate opera loving robot who put in so much time and energy to become a great opera singer robot, because the robot who she saw as her true love didn't look her way as she got better and didn't love her as much as she loved him. So throughout the boss fight, you get little snippets that show scenes of the turmoil that led to her current state of a broken heart. The music gets louder and dramatic and the opera robot's singing starts to turn into screaming and her attack patterns get more sporadic...so when you take a second to connect the dots with this story that's unraveling in front of you, it's emotionally challenging to end her, as on one hand it sucks that she went through this experience and you can end her suffering. But on the other hand, I wished I was able to help

her move on and let her know that there are and will be others who'll recognize and love her and her talents."

Through the detailed retelling of this particular scene using game design elements ("textual features") and alternative perspectives ("internalizing a point of view/not projecting their own"), this participant was not only more likely to identify with this character to a degree, but also faced an emotional challenge that influenced the way they experienced the game.

Building off of identification, with any relationship, there is a degree of emotional attachment. As Lewis et al. [36] states, emotional attachment can encompass a variety of an individual's feelings that can influence the connection a player has with character(s). Bopp et al. [8] conducted a study that researched how emotional qualities may categorize the type of attachment players may have with video game characters. They characterized seven distinct emotional attachment types: "Cool and Capable," "Respected Nemesis," "Admired Paragon," "Crush," "Concern for one's Protege," "Sympathetic Alter Ego," and "Trusted Close Friend".

Out of the seven types, Bopp et al. [8] discovered that three distinct emotional attachment types were consistently associated with PCs: "Admired Paragon", "Cool and Capable", and "Sympathetic Alter Ego". Bopp et al. [8] define "Admired Paragon" as characters that, "embodied virtues the participants aspired to be themselves... as role models to whom they look up to". "Cool and Capable" is defined as "feeling empowered, amused, and excited about a character's in-game capabilities". Lastly, "Sympathetic Alter Ego", which they characterize as when participants "felt sympathetic towards characters, because they felt they shared similar personal experiences with them".

One of our participants shared their experience when playing *Red Dead Redemption 2* as the PC, Arthur Morgan:

"I knew going into the game that there was an emotional storyline but I was surprised by how invested in the characters I became. Having to try and hold the gang together and resolve Arthur's issues as he dealt with people outside the gang was tough. Arthur started out cold and hard but as the story progressed, I was to make choices that showed a softer side and ultimately decided how Arthur would meet his end either selfishly or sacrificing himself for others."

Despite already knowing the "emotional storyline", this participant was still able to become emotionally attached to the PC (Arthur Morgan) as they stated that it was difficult to "hold the gang together (NPCs) and resolve Arthur's issues..." especially when making choices that led to them to see and experience Arthur change from a cold-hearted to softer cowboy. Hence, based on the participant's response, we can note that this participant's emotional attachment to the PC did correlate with the definitions of the top three forms of emotional attachment ("Admired Paragon", "Cool and Capable", and "Sympathetic Alter Ego") with a PC.

Hence, identification and emotional attachment through decision-making and/or actions can lead to unification with characters, but also present emotional challenges through requiring the emotional intelligence to empathize with them.

5.2.3 Autonomy Above All. As seen in Figure 1, the highest levels of reflection deal with a lot of autonomy-focused game design patterns, to include *Emotional Decision-Making*, *Empowerment*, and *Consequences of Long Ago Actions*. Coincidentally, Dormann et al. [22] previously argued that these three patterns are the most crucial foundations of social-emotional learning in games. We then argue that these above patterns are shared among deeper levels of reflection (e.g. transformative and critical reflection). Additionally, we then argue that these patterns are necessary to induce emotional challenge in gameplay, if one were to target a reflective player experience and a social-emotional learning gaming experience. One can then also assume that *Transformative Reflection* and *Critical Reflection* are the levels in which higher levels of learning occur for players. This furthers our understanding of how to best design serious games and games for learning.

Our findings align with previous work emphasizing the importance of autonomy in player experience [1, 46, 50]. Players typically enjoy having autonomy and the freedom to make decisions as they please. Emotional immersion by itself is not enough to stimulate deep reflective player experiences. In the case of evoking emotional challenge, players must have the autonomy to be directly involved in game events and actively immersed in its emotionally complex themes. Whitby et al. [54] found that the intersection of narrative and game systems where players can actually perform the actions to make decisions as part of gameplay made for memorable perspective-challenging moments. Having the emotional themes of the game be passive and something to stumble upon by accident cannot guarantee that players will take away reflective meaning from gameplay.

To further an earlier example, *Sacrificial Action* is one of the potential patterns that address an action being taken after *Emotional Decision-Making*. Action requires player *Empowerment*, which is "[when players feel] they can affect the events and final outcome of a game" [22], and results in *Consequences of Long Ago Actions*. When a player has the autonomy to sacrifice some aspect of their experience to benefit someone/something else in the game, it is a form of empathic process that takes place. It can then be argued that player autonomy is important when designing games for empathy as it stimulates a lasting reflective impact on players.

While autonomy in games is clearly important and often sought out, it is often hard to design for "full" autonomy and therefore a somewhat rare experience. Regardless, there are ways that games can be evaluated for their autonomy affordances even without giving players complete autonomy.

An example of how a game utilized a subtle lack of autonomy to emphasize its dramatic narrative can be seen in the following excerpt from one of our participants:

"Largely personal trauma combined with a distinct lack of agency in that moment. My mother is an unreliable alcoholic. Conway's alcoholism is not subtle, it's a recurring thing that's brought up in the game. In previous scenes though you are given a choice whether or not to drink (or at least whether or not to focus on Conway's drinking) and I had always picked a different choice because I wasn't super interested in a

story or an experience that revolved around alcoholism. You have no choice here. Conway will drink the expensive whiskey and take the job. You have to watch him self-destruct and there's nothing that you can do about it. It was viscerally unpleasant to me. This moment is vital for the game's story. It's central to what Kentucky Route Zero is about. But I felt powerless and furious at the game for making me make that choice in the moment. It was the marriage of me always having that choice before only to have it ripped away from me in that moment combined with just my general experience, I guess."

The player clearly felt emotionally challenged by the game placing their PC down a route that they were explicitly trying to avoid, especially because it related to a situation that matched their personal life. This use of a play on autonomy acted as a trigger for the player to connect the game deeper to reality and reflect on a personal level.

Hence, we advocate that through affective game design patterns, the inclusion of and focus on player autonomy design helps form more meaningful emotionally challenging experiences. However, just as demonstrated in the previous example, one needs to be mindful when it comes to autonomy design, especially in certain moments of games that are meant to be emotionally powerful as it can greatly influence the message sent to players.

5.3 Multiplayer Emotional Challenges

Something we did not initially foresee was the significant representation of multiplayer online games in about 26% of our responses. We thought that nearly all of the accounts would be about single-player gaming experiences, as those games are primarily where emotional challenge design is more of the focus. However, there were many accounts that involved both collaborating with a team and competing against an opposing team (e.g. *League of Legends*). While many of the previously identified affective design patterns [22] were applicable to these scenarios, such as *Healing/Nurturing Others* or *Sacrificial Action*, we do argue that it would be worth pursuing the identification of additional new multiplayer-focused affective design patterns, given the domination of online multiplayer games today. An example response is as follows:

"In the game Ultima Online you can become a thief character. You must balance your emotions to be able to steal from friends and keep your identity a secret. The game doesn't require that but to be a good thief you must steal and it makes people very angry. So playing a good thief meant keep my character a secret and stealing from friends at times. It is a challenge to keep in mind this is a game and not let your [sense] of right and wrong keep you from being the best thief you can be."

In the excerpt above, there is added complexity to the gameplay, as it occasionally requires performing a bad deed against a real-life person. Our findings support the concept of social challenge and/or

demand [10, 11, 19], which is defined as "the extent to which a video game triggers an implicit or explicit response in the player to the presence of other social actors." We argue that there is an additional layer that could be explored regarding emotional challenges when specifically playing with and/or against real-life players.

Though multiplayer games, especially competitive ones, often have the dynamic of adrenaline-filled emotions, it could still be useful to have affective design patterns and/or dynamics identified such as *Cooperation* or even *Revenge*. Such patterns could uncover how to best design for joint multiplayer reflective experiences, which is underexplored.

Overall, we argue that it is possible to target reflective emotionally challenging experiences in games through our identified affective design patterns. We hope that game researchers and designers alike can use this knowledge to effectively develop and research games with complex themes, serious games, and games for entertainment.

5.4 Study Limitations and Future Research

As some of our participant responses took a more player-centric focus for emotional challenge, we realized that there is a limitation in the existing academic definitions of "emotional challenge". Emotion in games is a highly subjective experience [28], and therefore, future work in emotional challenge research could potentially explore the expansion of existing emotional challenge definitions to consider how specific characters, actions, or objects may resonate differently with different players. Similarly, terms in the current definition such as "captivating story" should be more clearly operationalized.

Regarding our quantitative findings, while our statistical results show that there is not much of a drastic numerical difference in emotional challenge ratings once one has reached the *Dialogical Reflection* level, one must also note that that may not be the case when a game is played at the level of hundreds or even thousands of players. So certain patterns, such as providing *Emotional decision-making* to players, may actually result in the deepest reflections (*Critical Reflection*) one gets out of their player base. However, future larger scale studies are necessary to delve more deeply into this question. Additionally, our smaller sample size when breaking participants up into reflective groups could also explain why there was not a strong separation once reaching the *Dialogical Reflection* level.

A survey is limited in terms of not being as qualitatively rich as interviews, where one can follow up on initial responses given by participants. Additionally, a survey is not able to capture the objective data that one could obtain by recording gameplay and then asking or listening to participants reflect on what they are experiencing. It also should be noted that the recall of emotional experiences is not the same as the actual experience of emotions. That is why, for future research, we aim to conduct an observation and interview-focused study, where we are able to engage with participants more closely.

We were also limited by the number of responses we were able to receive, especially given that the higher levels of reflection (transformative and critical reflection) are more rare and harder to obtain when asking players to reflect [41]. Therefore, obtaining a larger

amount of participants would still most likely yield a disproportionate number of lower and higher levels of reflection. However, given our N=53 is split into five smaller groups, our results should be viewed carefully. It would be interesting to have a larger sample size in future studies to see if the trend continues. Regardless, we were able to balance getting both the qualitative and quantitative data we needed for this current study through our survey.

6 CONCLUSION

There is growing interest in reflective player experience research. Previous work has found that players value both reflection and uncomfortable emotions of varying degrees when playing games. It is then clear that emotional challenge, or the task of presenting players with narrative conflict and complex themes in gameplay, has some impact on reflective player experience. Through survey analysis and applying the codes for levels of reflection and affective game design patterns, we answer how emotional challenge in games relates to reflection. Secondly, we answer what affective game design patterns are most important to stimulate reflective player experiences. We found that there is a significant difference and mean ranks increased across the five reflection groups, meaning that deeper reflections come with more emotionally-challenging experiences. Additionally, we found that autonomy-based and negatively-valenced game design patterns were most referenced in the higher-level reflections, which support previous work in the area. We use these findings to discuss implications for affective and emotional challenge game design.

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REFERENCES

- [1] Vero Vanden Abeele, Katta Spiel, Lennart Nacke, Daniel Johnson, and Kathrin Gerling. 2020. Development and validation of the player experience inventory: A scale to measure player experiences at the level of functional and psychosocial consequences. *International Journal of Human-Computer Studies* 135 (2020), 102370.
- [2] Rosa María Baños, Cristina Botella, Mariano Alcañiz, Víctor Liaño, Belén Guerrero, and Beatriz Rey. 2004. Immersion and emotion: their impact on the sense of presence. *Cyberpsychology & behavior* 7, 6 (2004), 734–741.
- [3] Roy F Baumeister, Ellen Bratslavsky, Catrin Finkenauer, and Kathleen D Vohs. 2001. Bad is stronger than good. *Review of general psychology* 5, 4 (2001), 323–370.
- [4] Marit Bentvelzen, Paweł W Woźniak, Pia SF Herbes, Evropi Stefanidi, and Jasmin Niess. 2022. Revisiting Reflection in HCI: Four Design Resources for Technologies that Support Reflection. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 1 (2022), 1–27.
- [5] Max V Birk, Ioanna Iacovides, Daniel Johnson, and Regan L Mandryk. 2015. The false dichotomy between positive and negative affect in game play. In *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*. 799–804.
- [6] Staffan Björk and Jussi Holopainen. 2005. Games and design patterns. *The game design reader: A rules of play anthology* (2005), 410–437.
- [7] Julia Ayumi Bopp, Elisa D Mekler, and Klaus Opwis. 2016. Negative emotion, positive experience? Emotionally moving moments in digital games. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 2996–3006.
- [8] Julia Ayumi Bopp, Livia J Müller, Lena Fanya Aeschbach, Klaus Opwis, and Elisa D Mekler. 2019. Exploring emotional attachment to game characters. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. 313–324.
- [9] Julia Ayumi Bopp, Klaus Opwis, and Elisa D Mekler. 2018. “An Odd Kind of Pleasure” Differentiating Emotional Challenge in Digital Games. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–12.
- [10] Nicholas David Bowman, Joseph Wasserman, and Jaime Banks. 2018. Development of the video game demand scale. In *Video Games*. Routledge, 208–233.
- [11] Nicholas David Bowman, Rene Weber, Ron Tamborini, and John Sherry. 2013. Facilitating game play: How others affect performance at and enjoyment of video games. *Media Psychology* 16, 1 (2013), 39–64.
- [12] Virginia Braun and Victoria Clarke. 2012. *Thematic analysis*. American Psychological Association.
- [13] Virginia Braun and Victoria Clarke. 2021. One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative research in psychology* 18, 3 (2021), 328–352.
- [14] Virginia Braun and Victoria Clarke. 2022. Toward good practice in thematic analysis: Avoiding common problems and being a knowing researcher. *International Journal of Transgender Health* (2022), 1–6.
- [15] Bo Chang. 2019. Reflection in learning. *Online Learning* 23, 1 (2019), 95–110.
- [16] Jonathan Cohen. 2001. Defining identification: A theoretical look at the identification of audiences with media characters. *Mass communication & society* 4, 3 (2001), 245–264.
- [17] Tom Cole, Paul Cairns, and Marco Gillies. 2015. Emotional and functional challenge in core and avant-garde games. In *Proceedings of the 2015 annual symposium on computer-human interaction in play*. 121–126.
- [18] Mihály Csikszentmihályi, Sami Abuhamdeh, and Jeanne Nakamura. 2014. *Flow. In Flow and the foundations of positive psychology*. Springer, 227–238.
- [19] Alena Denisova, Paul Cairns, Christian Guckelsberger, and David Zendle. 2020. Measuring perceived challenge in digital games: Development & validation of the challenge originating from recent gameplay interaction scale (CORGIS). *International Journal of Human-Computer Studies* 137 (2020), 102383.
- [20] Claire Dormann and Robert Biddle. 2008. Understanding game design for affective learning. In *Proceedings of the 2008 Conference on Future Play: Research, Play, Share*. 41–48.
- [21] Claire Dormann, Jennifer R Whitson, and Robert Biddle. 2011. Computer games for affective learning. In *Handbook of research on improving learning and motivation through educational games: Multidisciplinary approaches*. IGI Global, 261–282.
- [22] Claire Dormann, Jennifer R Whitson, and Max Neuvians. 2013. Once more with feeling: Game design patterns for learning in the affective domain. *Games and Culture* 8, 4 (2013), 215–237.
- [23] Robert Elliott, Arthur C Bohart, Jeanne C Watson, and Leslie S Greenberg. 2011. Empathy. *Psychotherapy* 48, 1 (2011), 43.
- [24] Rowanne Fleck and Geraldine Fitzpatrick. 2010. Reflecting on reflection: framing a design landscape. In *Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction*. 216–223.
- [25] Chad Phoenix Rose Gowler and Ioanna Iacovides. 2019. “Horror, guilt and shame”—Uncomfortable Experiences in Digital Games. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. 325–337.
- [26] Michael James Heron and Pauline Helen Belford. 2015. All of your co-workers are gone: story, substance, and the empathic puzzler. *Journal of games criticism* 2, 1 (2015).
- [27] Robyn Hromek and Sue Roffey. 2009. Promoting Social and Emotional Learning With Games: “It’s Fun and We Learn Things”. *Simulation & gaming* 40, 5 (2009), 626–644.
- [28] Katherine Isbister. 2016. *How games move us: Emotion by design*. MIT Press.
- [29] Fengfeng Ke. 2016. Designing and integrating purposeful learning in game play: A systematic review. *Educational Technology Research and Development* 64 (2016), 219–244.
- [30] Rilla Khaled. 2018. Questions over answers: Reflective game design. *Playful disruption of digital media* (2018), 3–27.
- [31] Kristian Kiili. 2005. Digital game-based learning: Towards an experiential gaming model. *The Internet and higher education* 8, 1 (2005), 13–24.
- [32] Raph Koster. 2013. *Theory of fun for game design*. O’Reilly Media, Inc.
- [33] Max Kreminski, Ben Samuel, Edward Melcer, and Noah Wardrip-Fruin. 2019. Evaluating AI-based games through retellings. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, Vol. 15. 45–51.
- [34] Petri Lankoski and Staffan Björk. 2007. Gameplay design patterns for social networks and conflicts. In *Proc. Computer Game Design and Technology Workshop 2007*.
- [35] Nicole Lazzaro. 2008. The four fun keys. In *Game Usability: Advancing the Player Experience*. CRC press.
- [36] Melissa L Lewis, René Weber, and Nicholas David Bowman. 2008. “They may be pixels, but they’re MY pixels.” Developing a metric of character attachment in role-playing video games. *CyberPsychology & Behavior* 11, 4 (2008), 515–518.
- [37] Victor Lim-Fei, Huey Ming Woo, and Ming Yew Lee. 2016. Serious games to develop social and emotional learning in students. In *Serious Games: Second Joint International Conference, JCSG 2016, Brisbane, QLD, Australia, September 26-27, 2016, Proceedings 2*. Springer, 3–12.
- [38] Holbrook Mahn and Vera John-Steiner. 2002. The gift of confidence: A Vygotskian view of emotions. *Learning for life in the 21st century: Sociocultural perspectives on the future of education* (2002), 46–58.

- [39] Tim Marsh and Brigid Costello. 2012. Experience in serious games: between positive and serious experience. In *Serious Games Development and Applications: Third International Conference, SGDA 2012, Bremen, Germany, September 26-29, 2012. Proceedings 3*. Springer, 255–267.
- [40] Barbara Martin and Charles Reigeluth. 2013. Affective education and the affective domain: Implications for instructional-design theories and models. *Instructional-design theories and models: A new paradigm of instructional theory 2*, 1992 (2013), 485.
- [41] Elisa D Mekler, Ioanna Iacovides, and Julia Ayumi Bopp. 2018. "A Game that Makes You Question..." Exploring the Role of Reflection for the Player Experience. In *Proceedings of the 2018 annual symposium on computer-human interaction in play*. 315–327.
- [42] Elisa D Mekler, Stefan Rank, Sharon T Steinemann, Max V Birk, and Ioanna Iacovides. 2016. Designing for emotional complexity in games: The interplay of positive and negative affect. In *Proceedings of the 2016 annual symposium on computer-human interaction in play companion extended abstracts*. 367–371.
- [43] John W Payton, Dana M Wardlaw, Patricia A Graczyk, Michelle R Bloodworth, Carolyn J Tompsett, and Roger P Weissberg. 2000. Social and emotional learning: A framework for promoting mental health and reducing risk behavior in children and youth. *Journal of school health* 70, 5 (2000), 179–185.
- [44] Xiaolan Peng, Jin Huang, Alena Denisova, Hui Chen, Feng Tian, and Hongan Wang. 2020. A palette of deepened emotions: exploring emotional challenge in virtual reality games. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–13.
- [45] Xiaolan Peng, Jin Huang, Linghan Li, Chen Gao, Hui Chen, Feng Tian, and Hongan Wang. 2019. Beyond horror and fear: Exploring player experience invoked by emotional challenge in VR games. In *Extended abstracts of the 2019 CHI conference on human factors in computing systems*. 1–6.
- [46] Richard M Ryan, C Scott Rigby, and Andrew Przybylski. 2006. The motivational pull of video games: A self-determination theory approach. *Motivation and emotion* 30, 4 (2006), 344–360.
- [47] Phoebe Sengers, Kirsten Boehner, Shay David, and Joseph 'Jofish' Kaye. 2005. Reflective design. In *Proceedings of the 4th decennial conference on Critical computing: between sense and sensibility*. 49–58.
- [48] Anjuman Shaheen, Panagiotis Fotaris, and Sanaz Fallahkhair. 2021. A Systematic Review of Using Reflective Design Features in Game-Based Learning. In *15th European Conference on Game Based Learning ECGBL*.
- [49] Rhea Sharma, Dominic Kao, and Edward Melcer. 2022. Exploring Relevance and Meaningfulness from Unexpected Learning Experiences in Commercial Video Games. *DiGRA* (2022).
- [50] Penelope Sweetser and Peta Wyeth. 2005. GameFlow: a model for evaluating player enjoyment in games. *Computers in Entertainment (CIE)* 3, 3 (2005), 3–3.
- [51] Weimin Toh and David Kirschner. 2023. Developing social-emotional concepts for learning with video games. *Computers & Education* 194 (2023), 104708.
- [52] Wouter Van den Hoogen, Karolien Poels, Wijnand IJsselstein, and Yvonne De Kort. 2012. Between challenge and defeat: Repeated player-death and game enjoyment. *Media Psychology* 15, 4 (2012), 443–459.
- [53] Gabriela Walker and Jeni Venker Weidenbenner. 2019. Social and Emotional Learning in the age of virtual play: technology, empathy, and learning. *Journal of Research in Innovative Teaching & Learning* 12, 2 (2019), 116–132.
- [54] Matthew Alexander Whitby, Sebastian Deterding, and Ioanna Iacovides. 2019. "One of the baddies all along" Moments that Challenge a Player's Perspective. In *Proceedings of the annual symposium on computer-human interaction in play*. 339–350.
- [55] Matthew Alexander Whitby, Ioanna Iacovides, and Sebastian Deterding. 2023. "Conversations with pigeons": Capturing Players' Lived Experience of Perspective Challenging Games. *Proceedings of the ACM on Human-Computer Interaction* 7, CHI PLAY (2023), 833–855.
- [56] Georgios N Yannakakis and Ana Paiva. 2014. Emotion in games. *Handbook on affective computing 2014* (2014), 459–471.